



NPTEL Course on “Computer System Optimization for ML”

Lecture #10



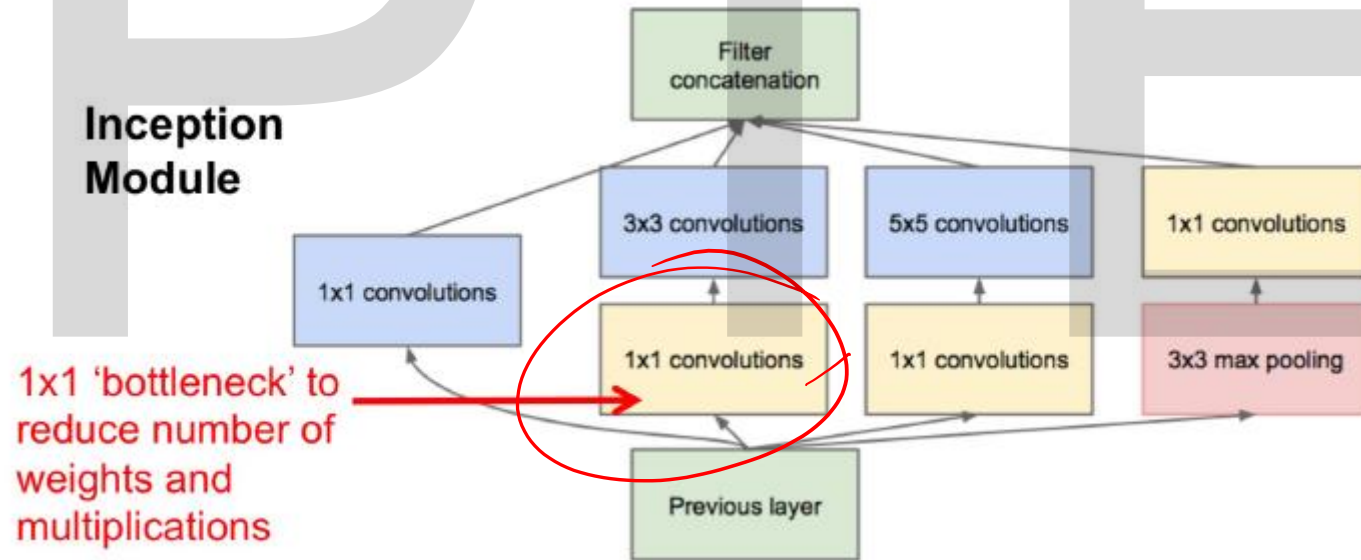
Agenda

- **Recap**
- **DenseNet**
- **Technique to Reduce FLOPS**

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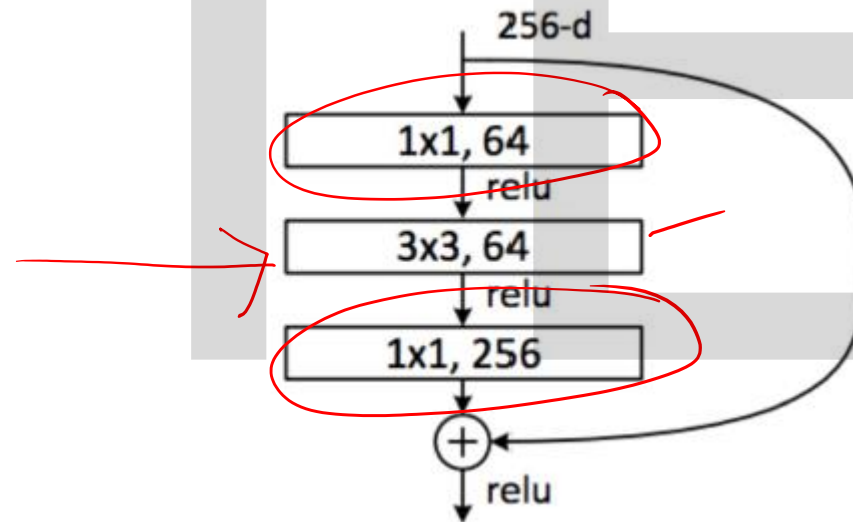
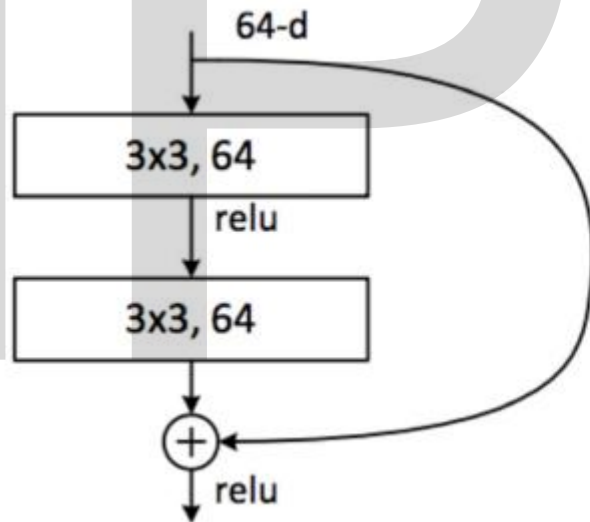
Inception V1

- Apply 1×1 before 'large' convolution filters
- Reduce weights such that the entire DNN can be trained on one GPU
- Number of multiplications reduced from 854M $\rightarrow$ 358M



RESNET: Use of 1x1 Filters

- Apply 1x1 filters to reduce computation and size
- Makes the network deeper (ResNet 34 → ResNet 50)



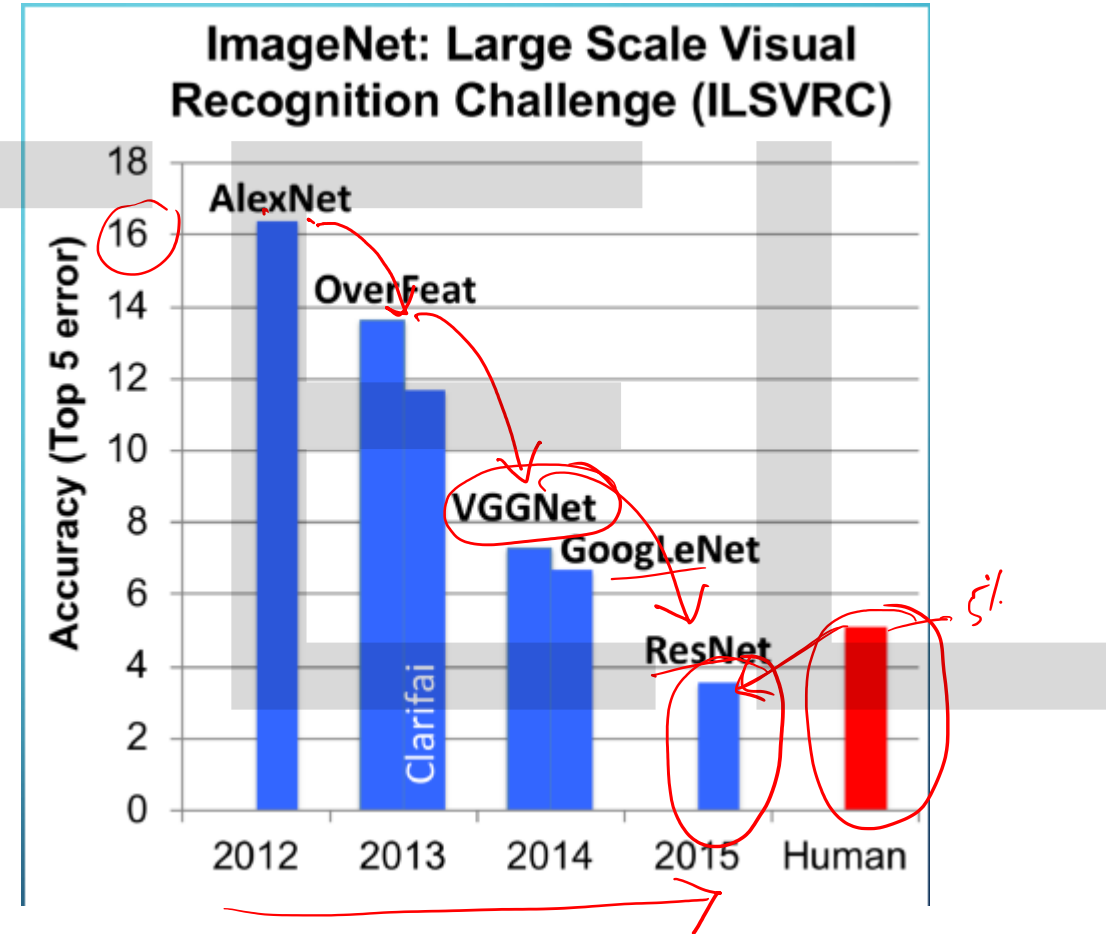
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- **DenseNet**
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List of Popular DNNs

- LeNet (1998) ✓
- AlexNet (2012) ✓
- OverFeat (2013) ✓
- VGGNet (2014) ✓
- GoogleNet (2014) ✓
- ResNet (2015) ✓



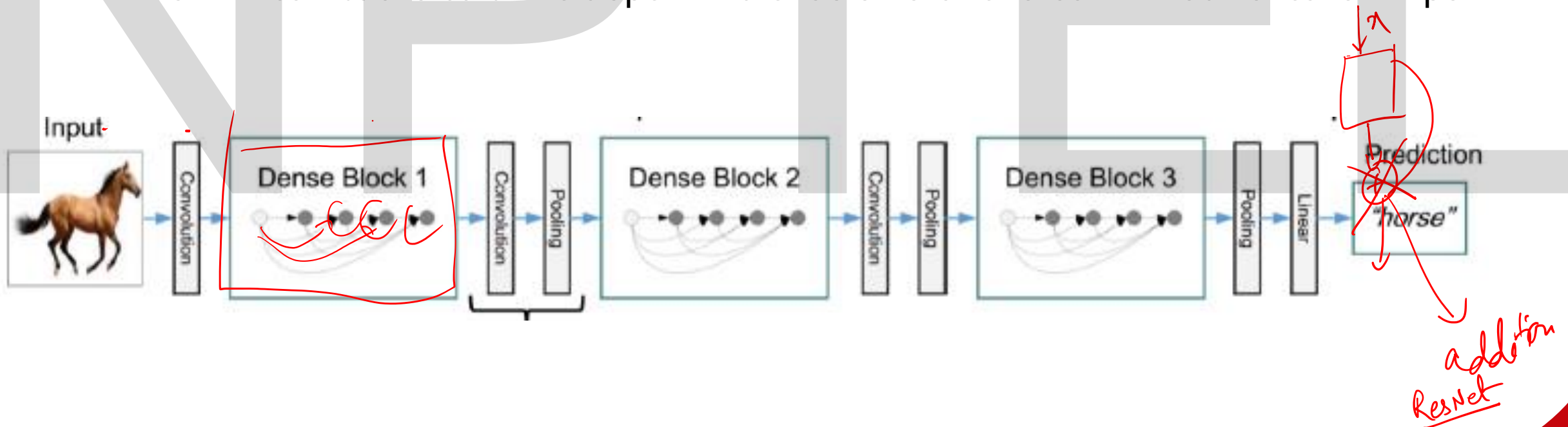
Summary of Popular DNNs

Metrics	LeNet-5	AlexNet	VGG-16	GoogLeNet (v1)	ResNet-50
Top-5 error	n/a	16.4	7.4	6.7	5.3
Input Size	28x28	227x227	224x224	224x224	224x224
# of CONV Layers	2	5	16	21 (depth)	49
Filter Sizes	5	3, 5, 11	3	1, 3, 5, 7	1, 3, 7
# of Channels	1, 6	3 - 256	3 - 512	3 - 1024	3 - 2048
# of Filters	6, 16	96 - 384	64 - 512	64 - 384	64 - 2048
Stride	1	1, 4	1	1, 2	1, 2
# of Weights	2.6k	2.3M	14.7M	6.0M	23.5M
# of MACs	283k	666M	15.3G	1.43G	3.86G
# of FC layers	2	3	3	1	1
# of Weights	58k	58.6M	124M	1M	2M
# of MACs	58k	58.6M	124M	1M	2M
Total Weights	60k	61M	138M	7M	25.5M
Total MACs	341k	724M	15.5G	1.43G	3.9G

DenseNet (Huan et al., CVPR 2017)

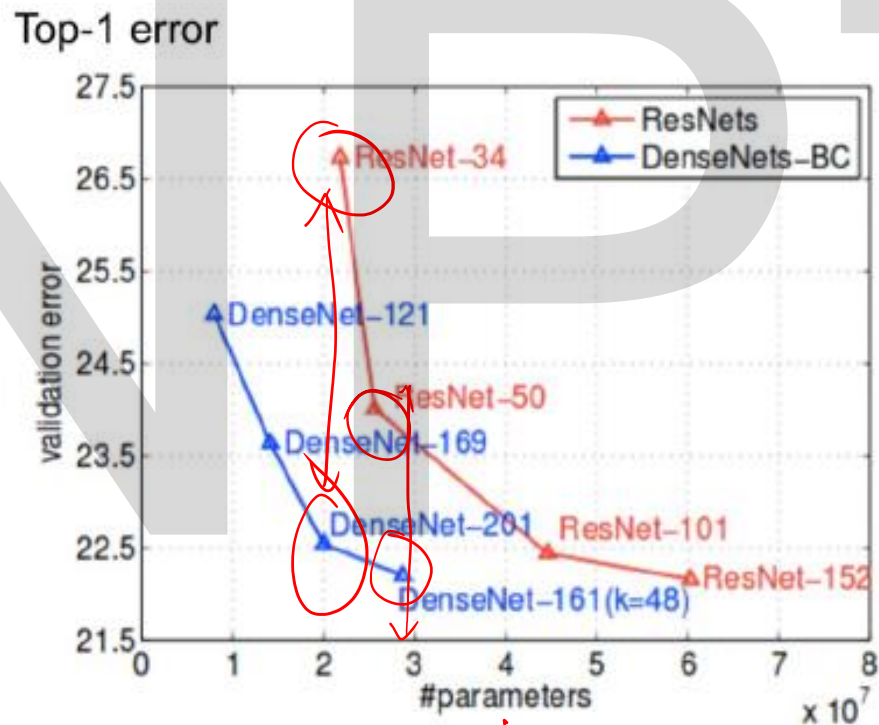
■ More Skip Connections

- Not only from previous layer but from many past layers to strengthen feature map propagation and feature reuse
- Features maps are concatenated rather than added
- Break into blocks to limit depth and thus size of the combined feature maps

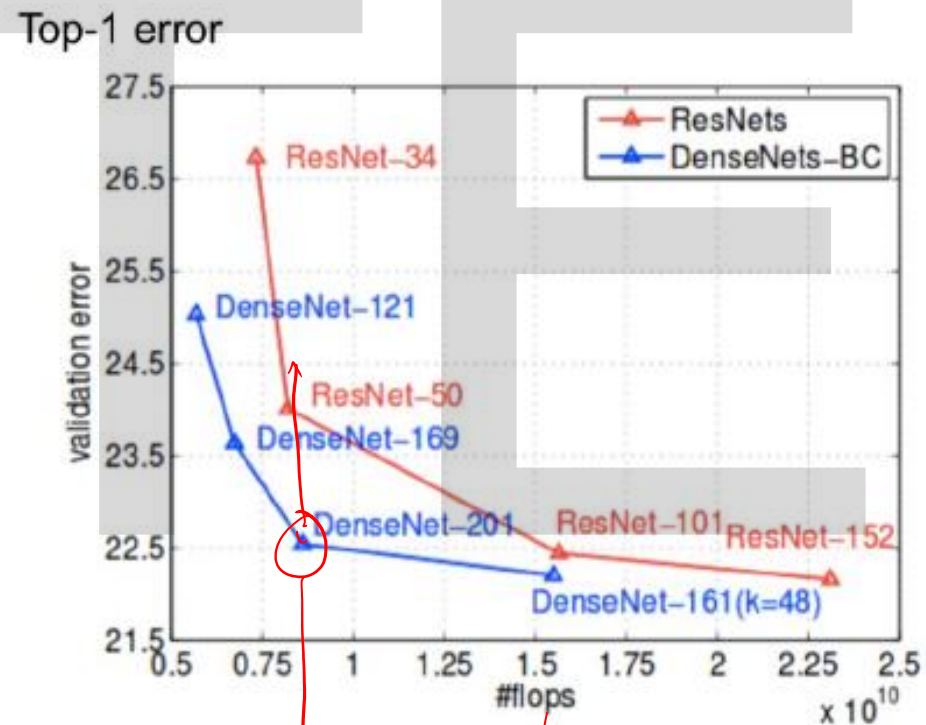


DenseNet

- Higher accuracy than ResNet with fewer weights and multiplications



↓
#weights



↓
0.80

↓
#MAC

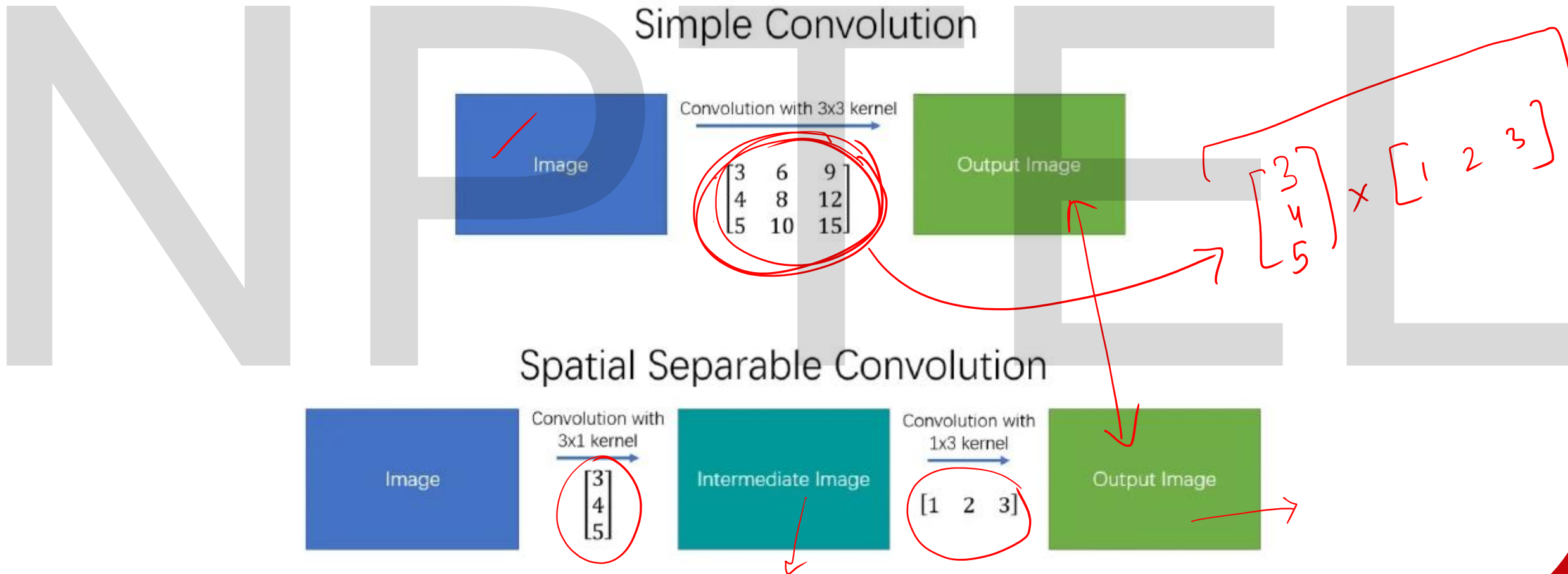
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- Recap
- DenseNet
- **Technique to Reduce FLOPS**

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Techniques to Reduce FLOPS (1)

- Floating Point Operations per Second (FLOPS)

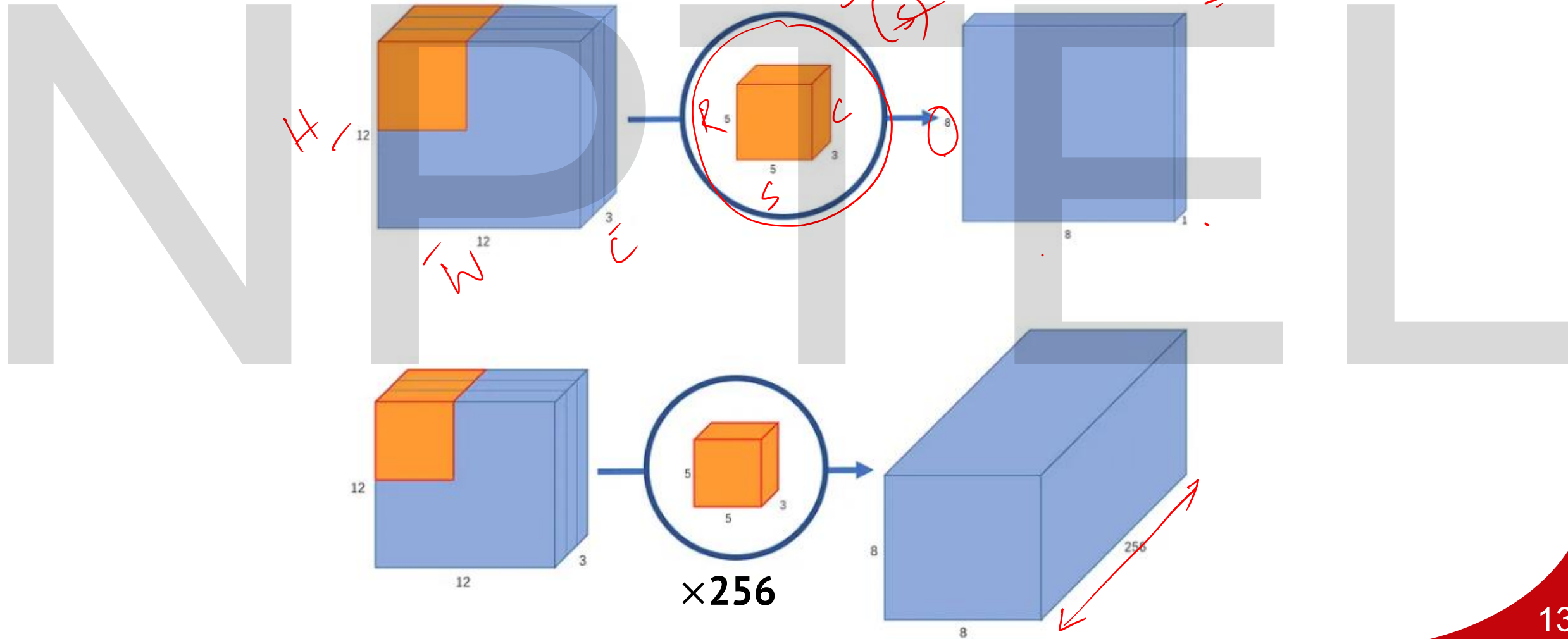


Drawbacks of Spatial Separable Convolution

- **No prior knowledge on filters**
 - They are being learnt
- **Hence not always possible to define matrix decompositions**
 - During training, the constraints need to be imposed

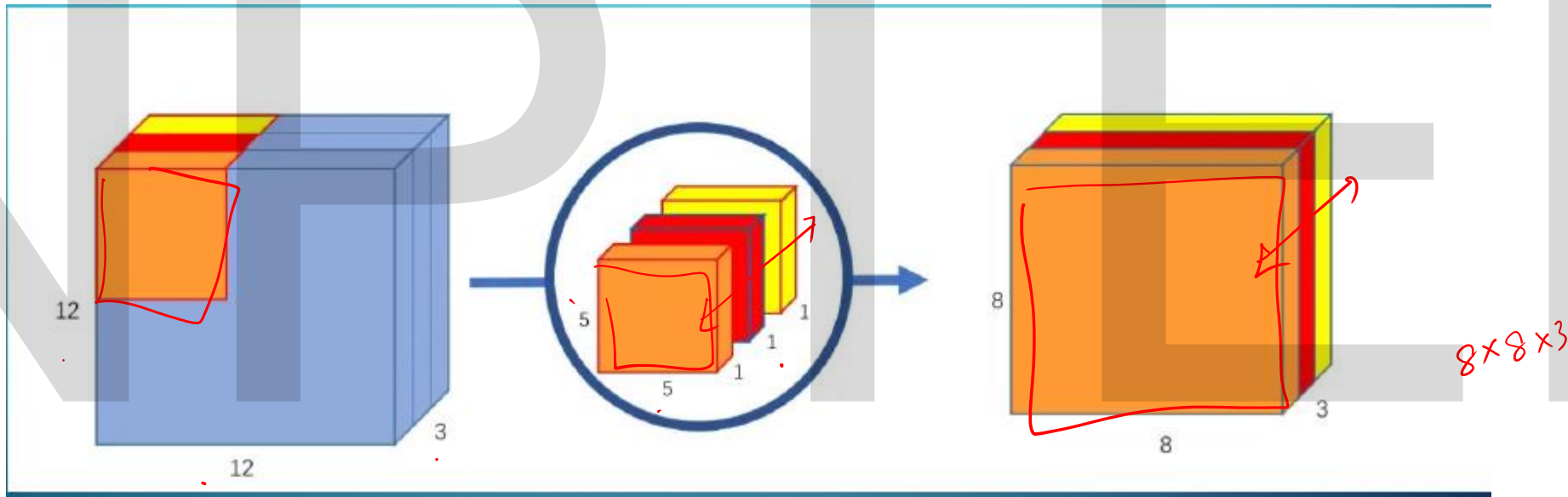
Techniques to Reduce FLOPS (2): Depth-wise Separable Conv

■ Regular Convolution



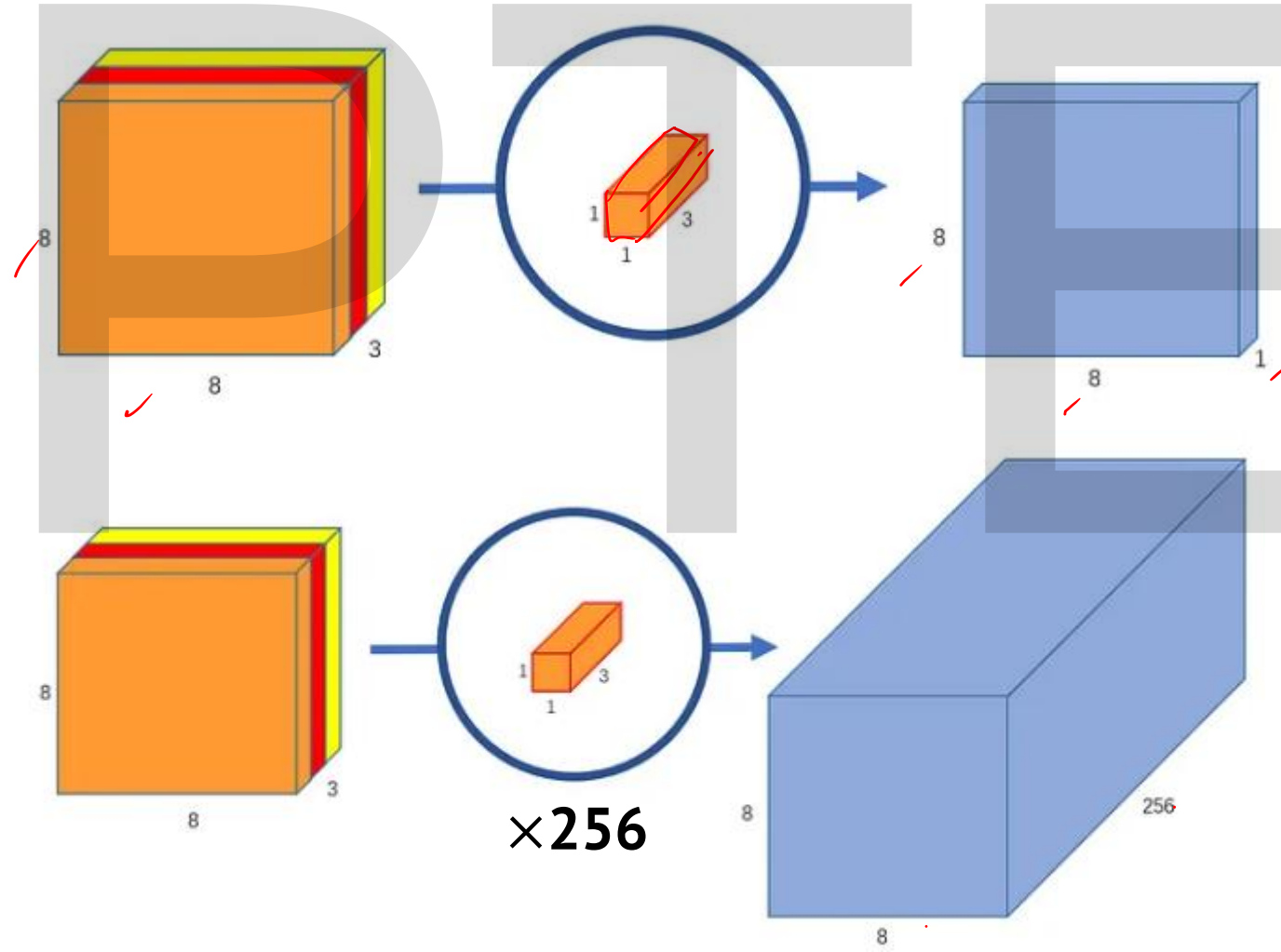
Depth-wise Separable Convolution

- Step-1: Depth-wise Convolution



Depth-wise Separable Convolution

■ Step-2: Point-wise Convolution



Advantage of Depth-wise Separable Convolution

- **Regular Convolution**

- $12 \times 12 \times 3 \otimes 5 \times 5 \times 3 \times 256 \rightarrow 8 \times 8 \times 256$

- **Depth-wise Separable Convolution**

- **Depth-wise:** $12 \times 12 \times 3 \otimes 5 \times 5 \times 1 \times 3 \rightarrow 8 \times 8 \times 3$

- **Point-wise:** $8 \times 8 \times 3 \otimes 1 \times 1 \times 3 \times 256 \rightarrow 8 \times 8 \times 256$

- **#Computations in Regular Convolution**

- $256 \ 5 \times 5 \times 3$ kernels move 8×8 times

- **#Multiplications:** $256 \times 5 \times 5 \times 3 \times 8 \times 8 = 1,228,800$

Advantage of Depth-wise Separable Convolution

- #Computations in Regular Convolution

- 256 $5 \times 5 \times 3$ kernels move 8×8 times

- #Multiplications: $256 \times 5 \times 5 \times 3 \times 8 \times 8 = 1,228,800$

- #Computation in Depth-wise Separable Convolution

- Depth-wise: $12 \times 12 \times 3 \otimes 5 \times 5 \times 1 \times 3 \rightarrow 8 \times 8 \times 3$

- #Multiplications: $3 \times 5 \times 5 \times 8 \times 8 = 4,800$

- Point-wise: $8 \times 8 \times 3 \otimes 1 \times 1 \times 3 \times 256 \rightarrow 8 \times 8 \times 256$

- #Multiplications: $256 \times 1 \times 1 \times 3 \times 8 \times 8 = 49,152$

- Total #Multiplications: 53,952

Intuition

- **Regular Convolution**

- Transform the original 12×12 image 256 times
- Every transformation uses $5 \times 5 \times 3 \times 8 \times 8 = 4800$ multiplications

- **In depth-wise separable convolution**

- Transform the original image only once ✓
- Then use a smaller kernel ✓



THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)

